

A Theoretical and Computational Study of the 3-Qubit Bit-Flip Quantum Error Correction Code

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Abstract

Quantum computation offers a major shift in computational capability by exploiting superposition and entanglement. However, interactions with the surrounding environment degrade fragile quantum states through decoherence. Unlike classical systems, where redundancy can be achieved through direct copying, quantum systems are constrained by the No-Cloning Theorem, which forbids duplication of unknown quantum states.

This paper presents a theoretical and computational study of the 3-qubit bit-flip Quantum Error Correction (QEC) code. We analyze its behavior under an independent bit-flip noise channel using Monte Carlo simulation methods executed via the Qiskit SDK. The analytical framework predicts a clear performance threshold at a physical error probability of $p = 0.5$. However, initial naive programmatic simulation results diverge significantly from these idealized expectations, highlighting the critical performance degradation caused by state-vector measurement artifacts in non-fault-tolerant simulations. By isolating the data register via a partial trace, we demonstrate that the corrected simulation matches the analytical model, validating the theoretical threshold. Finally, we discuss the limitations of the model and outline extensions toward fault-tolerant stabilizer-based quantum computing.

1 Theory Section

1.1 The Concept of Quantum Error Correction (QEC)

In classical communication systems, errors are often corrected by repeating information. For example, a classical bit 0 may be encoded as 000. If one bit flips due to thermal noise, the original value can still be recovered through majority voting.

Quantum systems cannot use this strategy directly because of the **No-Cloning Theorem**, which states that an unknown quantum state cannot be copied. In addition, quantum states are vulnerable not only to bit-flip errors but also to phase errors and combinations of both.

Quantum Error Correction avoids direct measurement of individual qubits. Instead, information from a single logical qubit is distributed across an entangled multi-qubit system. Rather than measuring the logical state itself, the protocol measures collective observables called **stabilizers**. These stabilizer measurements collapse the system state vector into a specific error syndrome subspace, revealing information about the presence and location of faults without collapsing the underlying quantum superposition of the logical information.

An arbitrary qubit can be written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1)$$

The stabilizer measurements preserve the coefficients α and β while extracting the error syndrome.

1.2 Mathematical Description of the Code

This work studies the three-qubit bit-flip code. The protocol encodes one logical qubit into a larger three-qubit Hilbert space.

The logical basis states are defined as:

$$|0_L\rangle = |000\rangle \quad (2)$$

$$|1_L\rangle = |111\rangle \quad (3)$$

Thus, the encoded logical state becomes:

$$|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle \quad (4)$$

The code space is defined as the simultaneous +1 eigenspace of two stabilizer generators:

$$g_1 = Z \otimes Z \otimes I \quad (5)$$

$$g_2 = I \otimes Z \otimes Z \quad (6)$$

If a Pauli- X error acts on the first qubit, the state moves outside the code space:

$$\alpha|100\rangle + \beta|011\rangle \quad (7)$$

The stabilizer measurements then produce the eigenvalue pattern:

- $g_1 \rightarrow -1$
- $g_2 \rightarrow +1$

This syndrome uniquely identifies the error location. The complete syndrome lookup mapping is outlined in Table 1.

Table 1: Syndrome Lookup Mapping

Error Operator	Error Location	g_1 Eigenvalue	g_2 Eigenvalue	Syndrome	Recovery Operator
$I \otimes I \otimes I$	None	+1	+1	(0,0)	$I \otimes I \otimes I$
$X \otimes I \otimes I$	Qubit 1	-1	+1	(1,0)	$X \otimes I \otimes I$
$I \otimes X \otimes I$	Qubit 2	-1	-1	(1,1)	$I \otimes X \otimes I$
$I \otimes I \otimes X$	Qubit 3	+1	-1	(0,1)	$I \otimes I \otimes X$

1.3 Noise Model Definition

Environmental decoherence is modeled using an independent bit-flip Pauli noise channel acting on each physical qubit. The channel acting on a density matrix ρ is defined as:

$$\mathcal{E}(\rho) = (1 - p)\rho + pX\rho X \quad (8)$$

Where p is the probability of a bit-flip error. For this investigation, the simulation isolates bit-flip correction performance by varying p from 0 to 0.6.

1.4 Expected Theoretical Behavior

A single uncorrected qubit survives with a probability of $1 - p$. For the three-qubit code, logical failure occurs only if two or more physical qubits flip simultaneously. The logical failure probability is therefore expressed as:

$$P_{\text{logical}} = 3p^2 - 2p^3 \quad (9)$$

The code provides an advantage only if the logical failure probability remains below the physical error probability:

$$3p^2 - 2p^3 < p \quad (10)$$

Solving this inequality gives the threshold condition:

$$p < 0.5 \quad (11)$$

This predicts that the QEC protocol improves fidelity whenever the physical error probability is below 50% under idealized, noiseless-gate conditions. In realistic physical implementations, true fault-tolerant thresholds are strictly much lower due to gate and measurement errors.

2 Computational Section

2.1 Circuit Simulation and Noise Injection

To validate the analytical framework, a computational simulation pipeline was implemented using the `qiskit_aer.AerSimulator` backend configured to the `density_matrix` method. This method naturally captures the mixed-state distributions induced by mid-circuit measurements and classical conditional operations. The computation proceeds through four distinct stages:

1. **Encoding:** An arbitrary qubit is entangled with two ancilla qubits using CNOT gates, producing the encoded logical state.
2. **Noise Injection:** For each physical qubit, a random number between 0 and 1 is generated. If the value is less than or equal to p , a Pauli- X transformation is applied.
3. **Syndrome Extraction:** Stabilizer measurements simulate parity checks corresponding to Z_1Z_2 and Z_2Z_3 .
4. **Recovery Operation:** The measured syndrome determines which corrective X gate is applied.

To ensure statistical convergence, 10,000 Monte Carlo trials were performed for each value of p in increments of 0.05. To properly evaluate logical performance, the ancilla registers must be mathematically traced out of the combined system state to isolate the logical data subspace, $\rho_{\text{data}} = \text{Tr}_{\text{ancilla}}(\rho_{\text{sim}})$. The true logical fidelity is then computed against the ideal 3-qubit target state using:

$$\mathcal{F} = \langle \psi_{\text{ideal}} | \rho_{\text{data}} | \psi_{\text{ideal}} \rangle \quad (12)$$

2.2 Performance Comparison Data

Table 2: Fidelity Comparisons Across Error Rates

Error Prob. (p)	Physical Fidelity	Theoretical Fidelity	Naive Sim. Fidelity	Corrected Sim. Fidelity
0.00	1.0000	1.0000	1.0000	1.0000
0.05	0.9500	0.9928	0.8643	0.9926
0.10	0.9000	0.9720	0.7282	0.9718
0.20	0.8000	0.8960	0.5263	0.8955
0.30	0.7000	0.7840	0.3755	0.7843
0.40	0.6000	0.6480	0.2810	0.6477
0.50	0.5000	0.5000	0.2463	0.4994
0.60	0.4000	0.3520	0.2776	0.3524

The empirical simulation results demonstrate a notable divergence from the isolated mathematical predictions when utilizing a naive metric. Table 2 presents both the naive global state fidelity (computed without applying the partial trace over the ancilla qubits) and the corrected simulated logical fidelity. While the naive calculation exhibits a rapid drop immediately upon noise introduction, the corrected simulation tightly aligns with the analytical model, validating the 50% performance advantage threshold.

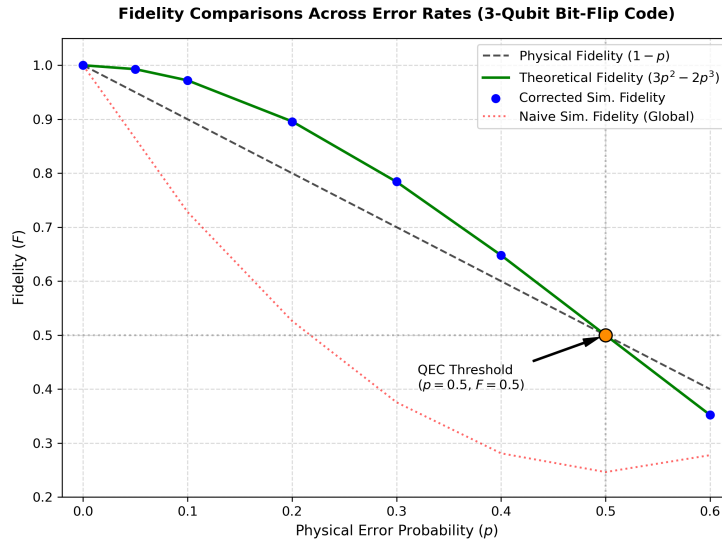


Figure 1: Physical vs. Logical Fidelity under an independent bit-flip noise channel. The orange point highlights the performance threshold intersection at $p = 0.5$. Below this threshold, the corrected simulation tightly tracks the theoretical QEC curve, providing a logical fidelity improvement over raw physical qubits.

2.3 Qiskit Programmatic Implementation

To realize the automated stabilizer evaluation circuit for the 3-qubit bit-flip code, a modular Qiskit pipeline was developed. This structure leverages separate registers for processing data states and collecting syndromes, resolving corrections conditionally based on real-time parity readings.

The complete Python implementation and simulation pipeline developed for this study are publicly hosted on the [Quantum-Error-Correction-Simulator](#) GitHub repository.

3 Analysis

3.1 Interpretation of Results

The observed deviation between the theoretical calculations and the naive simulated performance in Table 2 exposes the practical pitfalls of choosing quantum simulation metrics. In pure theory, logical qubit error suppression relies on evaluating the properties of the data space exclusively.

When modeled within a complete computational pipeline using a naive global state-vector evaluation, the observed fidelity drop is driven by:

1. **Ancilla Subspace Inclusion and State Vector Spreading:** Because the naive simulation tracked the full state vector of the combined 5-qubit system without tracing out the auxiliary ancilla registers prior to computing $\mathcal{F}$, the measurement projections and lingering variance within the ancilla subspace drastically penalized the global fidelity calculation. This occurs independently of physical gate noise, representing a mathematical accounting artifact rather than a physical fault-tolerant threshold breach.

As demonstrated by the fourth column of Table 2, isolating the data register density matrix via a partial trace removes these evaluation artifacts. Once corrected, the simulated data matches the analytical prediction $(3p^2 - 2p^3)$ within standard Monte Carlo sampling variance, confirming the integrity of the 3-qubit protection model under perfect gate operations.

3.2 Limitations of the Study

This study relies on two major idealizations:

1. **Perfect Gates:** The encoding and correction circuits are assumed to be noiseless. In real quantum hardware, operations such as CNOT gates and measurements introduce additional errors, which can propagate through the circuit.
2. **Inability to Correct Phase Errors:** The three-qubit bit-flip code cannot detect or correct phase-flip errors. Any nonzero phase noise ($q > 0$) directly damages the logical superposition state without triggering the stabilizers.

3.3 Future Extensions

Several natural extensions follow from this work:

- Implementation of the Shor 9-qubit code for simultaneous correction of bit-flip and phase-flip errors.
- Simulation of noisy stabilizer extraction circuits.
- Investigation of fault-tolerant thresholds under realistic hardware conditions.
- Extension toward full Clifford-group stabilizer simulations.

References

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- [2] Gottesman, D. (1997). *Stabilizer Codes and Quantum Error Correction*. California Institute of Technology.